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A formal and reproducible framework for auditing observable behavior in qualitative biological network models

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Structurally similar qualitative biological networks can differ in reachable branching, whereas mechanistically different models may agree when internal events are hidden. We map Petri nets and logical models to finite reachable labeled transition systems and classify strong/weak bisimilarity, directional weak simulation, or non-comparability under a declared interface. We prove correctness and termination for finite systems. Six construction-level controls served only as software verification: weak bisimulation recovered 6/6 decisions, and Python and mCRL2 agreed on 42 strong/weak decisions. The central case asks where encoded animal/human and *Arabidopsis* DNA-damage responses retain common control behavior. Their unchanged structures scored 0.9976 by PN-GDDA. Coarse and pathway-resolved interfaces were weakly bisimilar; a mechanism-resolved interface distinguishing mammalian apoptosis from plant SMR and differentiation/endoreduplication became non-comparable ($d_6 = 0.353$). This locates a model-derived boundary between shared function and mechanism-specific behavior without implying organism-level equivalence. In an exploratory HPN-DREAM analysis, nine directional relations were retained without ordinal encoding; associations and the CASPOTS native-context advantage were inconclusive (exact $p = 0.25$). Structural and behavioral comparisons therefore provide complementary, interface-conditional evidence.

Keywords: Qualitative biological networks; Petri nets; labeled transition systems; weak bisimulation; model checking; Boolean networks.

1. Introduction

Qualitative biological networks make regulatory and signaling assumptions executable when kinetic information is unavailable or deliberately abstracted. Models intended to represent related functions may nevertheless come from different organisms, formalisms, curation choices, or update semantics. A useful comparison must therefore ask not only whether their graphs look alike, but whether their reachable executions preserve the biological events and choices relevant to a stated question. Petri nets represent causality, conflict, and concurrency explicitly, whereas logical models encode discrete regulatory rules.^{1–4}

Structural similarity does not determine observable executable behavior. Graphlet methods sensitively compare local motifs; PN-GDDA, for example, uses

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151 directed bipartite Petri-net graphlets and 592 published orbit slots.⁵ By design, that score does not encode transition labels, initial marking, or recursively reachable branching. Conversely, internal refinement can alter topology without changing behavior visible at a chosen interface. Structural and behavioral comparisons consequently answer different, complementary questions.

This distinction is biological as well as technical. Animal/human and plant DNA-damage responses share high-level functions such as lesion sensing, checkpoint activation, repair, and recovery, but implement regulation and terminal responses differently. Mammalian networks include p53-associated checkpoint and apoptotic programs, whereas *Arabidopsis thaliana* uses the plant-specific SOG1 network, WEE1 and SMR checkpoint regulators, and damage-associated differentiation or endoreduplication responses.^{6–11} The resulting modeling question is resolution-dependent: at what biological level do the two encoded control structures preserve the same observable behavior, and when do mechanism-specific outcomes make that correspondence fail? The set of observable labels, denoted by Σ , is therefore treated as a biological hypothesis to be declared and tested rather than a neutral software option.

Linear traces add event order but omit where alternatives become committed. Two models can generate the same finite sequences while resolving a biological choice at different states. Branching-time equivalences retain that information, and simulation preorders report asymmetric containment when every behavior of one model can be matched by another. Recent formal work on asynchronous Boolean networks likewise highlights the role of reachability and branching structure in phenotypic analysis.¹²

Let N_1 and N_2 be executable labeled models and let Σ be a declared observable interface. We formulate the comparison as

$$(N_1, N_2, \Sigma) \mapsto (\mathcal{L}(N_1), \mathcal{L}(N_2)) \mapsto C_\Sigma(\mathcal{L}(N_1), \mathcal{L}(N_2)), \quad (1.1)$$

where $\mathcal{L}(N_i)$ is the finite reachable labeled transition system and C_Σ returns the strongest supported class among strong bisimilarity, weak bisimilarity, mutual weak simulation, either one-way weak simulation, and non-comparability. We call the reporting priority *graded*: a stronger relation suppresses weaker implied labels. This is an implication-aware classifier, not an algebraic grading. Structural and truncated-trace measures remain attached diagnostics rather than substitutes for C_Σ .

The formal relations are standard, but their application is made auditable. We define the observable interface and simulation orientation, prove greatest-fixed-point correctness for the executed relation-deletion algorithm, and establish the hierarchy supporting the classifier. Controlled silent refinement and minimal counterexamples separate structure, traces, and branching. Construction-level cases, mCRL2, and an independently coded attacker–defender game verify implementation behavior; they are not presented as biological validation.

The biological application is organized around the GIM DNA-damage case. Three interfaces fixed from the model definitions and literature range from coarse

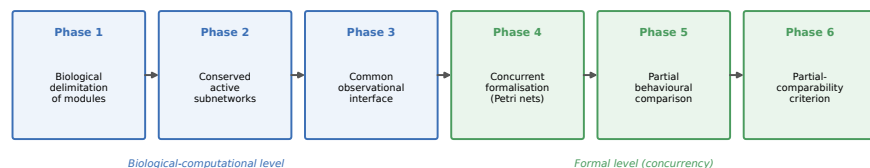


Fig. 1. Six-stage workflow. Stages 1–3 delimit the model-comparison question and observational interface; stages 4–6 construct executable Petri nets, compare their reachable behavior, and assign a model-level relation.

function to mechanism-resolved terminal fate; the model structures are held constant. A secondary RCD case illustrates directional containment. Public GINsim controls test SBML-qual/GINML ingestion and update semantics, not biological truth.^{13–16} Finally, nine HPN-DREAM/CASPOTS pair-condition observations show how formal outputs can be confronted with held-out perturbation data, but the analysis is explicitly exploratory.^{17,18}

The contributions are:

- (i) a formal bioinformatics framework that audits two executable qualitative biological models under a declared interface;
- (ii) mathematical guarantees for the implemented classifier, together with properties and counterexamples distinguishing structure, traces, and branching;
- (iii) independent formal/software verification through six controlled cases, mCRL2, 67,600 attacker-defender comparisons, regression tests, and observed computational scaling; and
- (iv) a multi-interface GIM analysis that separates high-level encoded correspondence from organism-specific terminal behavior, with supporting public model controls and an exploratory held-out HPN analysis.

Throughout, four evidence layers are kept separate: formal correctness, software/implementation verification, support for executable model construction and ingestion, and independent biological evidence or interpretation. PN-GDDA and the structural profile summarize topology; traces retain order but not the location of choices; LTS-GDA retains labeled local motifs; bisimulation and simulation match recursive branching. None of these formal outputs alone establishes that a curated model is experimentally valid.

The operational sequence is summarized in Fig. 1. Biological question and interface declaration precede formal encoding, reachability, and classification, making the abstraction assumptions inspectable before an equivalence calculation is interpreted.

2. Formal framework for observable biological-model comparison

2.1. Labeled Petri nets and induced transition systems

Definition 1 (Labeled place/transition net). A labeled place/transition net is a tuple

$$N = (P, T, \text{pre}, \text{post}, \ell, m_0), \quad (2.1)$$

where P and T are finite sets of places and transitions, $\text{pre}, \text{post} : T \rightarrow \mathbb{N}^P$ are the input and output multisets, $m_0 \in \mathbb{N}^P$ is the initial marking, and $\ell : T \rightarrow \Sigma \cup \{\tau\}$ assigns either an observable action in the declared interface Σ or the internal action τ .

Under standard interleaving semantics, t is enabled at m when $\text{pre}(t) \leq m$ componentwise; firing it changes m to $m - \text{pre}(t) + \text{post}(t)$. The implementation uses a worklist to enumerate reachable markings explicitly. A state cap and a token bound make failure to establish a finite state space explicit rather than silently truncating the relation calculation. All curated Petri nets used in Section 4 are safe, but the definitions below apply to every finite reachable system.

Definition 2 (Finite labeled transition system). A finite labeled transition system (LTS) is

$$L = (S, \Sigma_\tau, \longrightarrow, s_0), \quad \Sigma_\tau = \Sigma \cup \{\tau\}, \quad (2.2)$$

where S is a finite state set, $s_0 \in S$, and $\longrightarrow \subseteq S \times \Sigma_\tau \times S$. We write $s \xrightarrow{a} s'$ for $(s, a, s') \in \longrightarrow$. The reachable semantics of the net in Definition 1 is denoted by $\mathcal{L}(N)$.

2.2. Observable and weak transitions

Definition 3 (Silent closure and weak target sets). For $s \in S$, let

$$\text{Cl}_\tau(s) = \{u \in S : s \xrightarrow{\tau^*} u\}. \quad (2.3)$$

For $a \in \Sigma$, define

$$W_a(s) = \{v : \exists u, w \quad s \xrightarrow{\tau^*} u \xrightarrow{a} w \xrightarrow{\tau^*} v\}, \quad W_\tau(s) = \text{Cl}_\tau(s). \quad (2.4)$$

We write $s \xRightarrow{a} v$ when $v \in W_a(s)$.

The convention $W_\tau = \text{Cl}_\tau$ is important: a direct silent move on one side may be matched by zero or more silent moves on the other. It is also the convention implemented by the Python engine and used in the AUT exports checked by mCRL2.

2.3. Simulation and bisimulation

Let $L_i = (S_i, \Sigma_\tau, \longrightarrow_i, s_i^0)$ for $i \in \{1, 2\}$.

Definition 4 (Weak simulation). A relation $R \subseteq S_1 \times S_2$ is a weak simulation from L_1 to L_2 if, whenever $(x, y) \in R$ and $x \xrightarrow{a}_1 x'$, there is $y' \in W_a^2(y)$ such that $(x', y') \in R$. We write

$$L_1 \preceq_w L_2 \quad (2.5)$$

if (s_1^0, s_2^0) belongs to some such relation. Thus the right-hand system L_2 simulates the left-hand system L_1 ; the left-hand behavior is contained in the right-hand behavior under the interface.

Definition 5 (Weak and strong bisimulation). A relation $R \subseteq S_1 \times S_2$ is a weak bisimulation if it satisfies the condition in Definition 4 and, for every $(x, y) \in R$ and $y \xrightarrow{a}_2 y'$, there is $x' \in W_a^1(x)$ with $(x', y') \in R$. The initial systems are weakly bisimilar, written $L_1 \sim_w L_2$, when their initial pair belongs to such a relation. Strong bisimulation, written $L_1 \sim_s L_2$, replaces every weak target set $W_a^i(x)$ by the direct target set $\{x' : x \xrightarrow{a}_i x'\}$, including for $a = \tau$.

These are standard branching-time relations.^{19,20} The explicit orientation in (2.5) prevents the common ambiguity between “simulates” and “is simulated by” in one-way results.

2.4. Weak trace language

Definition 6 (Exact and truncated weak traces). The finite weak observable language of L is

$$L_w(L) = \{a_1 \cdots a_r \in \Sigma^* : s_0 \xRightarrow{a_1} \cdots \xRightarrow{a_r} s \text{ for some } s \in S\}. \quad (2.6)$$

Its depth- k restriction is $L_{\leq k}(L) = \{u \in L_w(L) : |u| \leq k\}$. The diagnostic distance is

$$d_k(L_1, L_2) = 1 - \frac{|L_{\leq k}(L_1) \cap L_{\leq k}(L_2)|}{|L_{\leq k}(L_1) \cup L_{\leq k}(L_2)|}. \quad (2.7)$$

The implementation computes the prefix-closed truncated sets in Definition 6. The statement $d_k = 0$ is therefore not identified with exact weak-trace equivalence. Exact weak-trace results in this study come only from mCRL2 or from a finite construction for which the complete language is enumerated analytically.

2.5. Graded behavioral classification

Definition 7 (Graded classifier). For finite L_1, L_2 over the declared interface Σ , define $C_\Sigma(L_1, L_2)$ by the following priority:

$$C_\Sigma(L_1, L_2) = \begin{cases} \text{strong,} & L_1 \sim_s L_2, \\ \text{weak,} & L_1 \not\sim_s L_2 \text{ and } L_1 \sim_w L_2, \\ \text{mutual,} & L_1 \not\sim_w L_2, L_1 \preceq_w L_2, L_2 \preceq_w L_1, \\ \text{left_in_right,} & L_1 \preceq_w L_2, L_2 \not\preceq_w L_1, \\ \text{right_in_left,} & L_2 \preceq_w L_1, L_1 \not\preceq_w L_2, \\ \text{noncomparable,} & \text{otherwise.} \end{cases} \quad (2.8)$$

The output is a reporting class, not a newly proposed equivalence relation.

Proposition 1 (Exclusive reporting). *The classifier in Definition 7 assigns exactly one class to every ordered pair of finite LTSs.*

Proof. The first two cases are selected before any simulation-only case. If neither is selected, the two simulation predicates have exactly four Boolean combinations: both true, only the first true, only the second true, or both false. These are the last four cases of (2.8), respectively. The cases are exhaustive and disjoint by their stated guards. $\square$

2.6. Fixed-point computation and formal properties

For $a \in \Sigma_\tau$, let $D_a^i(x) = \{x' : x \xrightarrow{a}_i x'\}$ and $U = S_1 \times S_2$. Define operators on $\mathcal{P}(U)$ by

$$\Phi_{\preceq}(R) = \{(x, y) \in U : \forall a, x' \in D_a^1(x) \exists y' \in W_a^2(y), (x', y') \in R\}, \quad (2.9)$$

$$\Phi_w(R) = \{(x, y) \in \Phi_{\preceq}(R) : \forall a, y' \in D_a^2(y) \exists x' \in W_a^1(x), (x', y') \in R\}, \quad (2.10)$$

$$\begin{aligned} \Phi_s(R) = \{(x, y) \in U : \forall a, x' \in D_a^1(x) \exists y' \in D_a^2(y), (x', y') \in R, \\ \forall a, y' \in D_a^2(y) \exists x' \in D_a^1(x), (x', y') \in R\}. \end{aligned} \quad (2.11)$$

Theorem 1 (Correctness of relation deletion). *For each $\Phi \in \{\Phi_{\preceq}, \Phi_w, \Phi_s\}$, the operator is monotone. The implemented algorithm starts with $R = U$, repeatedly deletes any currently examined pair $z \notin \Phi(R)$, and stops after a complete pass with no deletion. On finite LTSs it terminates and returns the greatest fixed point $\nu\Phi$. Consequently, testing membership of (s_1^0, s_2^0) computes, respectively, weak simulation in the orientation of Definition 4, weak bisimilarity, and strong bisimilarity.*

Proof. If $R \subseteq Q$, every witness pair in R remains available in Q ; hence $\Phi(R) \subseteq \Phi(Q)$ for each operator. Every deletion strictly decreases the finite relation, so termination requires at most $|U|$ deletions.

Let $G = \nu\Phi$. The invariant $G \subseteq R$ holds initially. If it holds before a deletion and $z \in G$, then $z \in \Phi(G) \subseteq \Phi(R)$, so z cannot be deleted. Thus G remains a

subset of the terminal relation R_* . A final pass without deletion gives $R_* \subseteq \Phi(R_*)$. Conversely, each $z \notin R_*$ was deleted from some R_z with $z \notin \Phi(R_z)$. Since $R_* \subseteq R_z$, monotonicity implies $\Phi(R_*) \subseteq \Phi(R_z)$ and therefore $z \notin \Phi(R_*)$. Hence $R_* = \Phi(R_*)$. Because R_* is a fixed point, $R_* \subseteq G$; together with the invariant this yields $R_* = G$. The three operators are exactly the transfer clauses of Definitions 4 and 5. $\square$

Remark 1 (Resource bound). Writing $n_i = |S_i|$, the candidate relation stores $O(n_1 n_2)$ pairs and permits at most $n_1 n_2$ deletions and $n_1 n_2 + 1$ passes. Each pass scans current pairs, outgoing moves, and matching direct or weak targets. Silent closures are cached; visible weak targets are constructed on demand. We therefore report these parameters instead of claiming an unjustified tight time bound.

Proposition 2 (Hierarchy of behavioral relations). *For finite LTSs under Definitions 3–6,*

$$L_1 \sim_s L_2 \implies L_1 \sim_w L_2 \implies (L_1 \preceq_w L_2 \wedge L_2 \preceq_w L_1) \implies L_w(L_1) = L_w(L_2). \quad (2.12)$$

The converses do not hold in general.

Proof. A direct match is a permitted weak match, and each direction of a weak bisimulation is a weak simulation. If $L_1 \preceq_w L_2$, induction along any finite concrete path matches observable moves by $\tau^* a \tau^*$ and silent moves by τ^* , preserving related successors. Erasing τ gives $L_w(L_1) \subseteq L_w(L_2)$; the reverse simulation gives equality. Proposition 3 and Example 1 refute the first and last converses. For the middle converse, compare $a.(b+c) + a.b$ with $a.(b+c)$: each simulates the other, but the $a.b$ successor cannot be bisimilar to a state offering both b and c . $\square$

Proposition 3 (Controlled silent edge refinement). *Let L contain $p \xrightarrow{a} q$ with $a \in \Sigma$ and $p \neq q$. Construct L' by copying all original states and transitions except this edge, and replacing it by*

$$\bar{p} \xrightarrow{a} r \xrightarrow{\tau} \bar{q}, \quad (2.13)$$

where r is fresh and has no other incident behavior. Then $L \sim_w L'$. Repeated refinements of this form also preserve weak bisimilarity. If p and $\bar{p}$ are initial, the selected edge is the only a -edge from $\bar{p}$ in L' , and every a -successor of p in L has no outgoing τ transition, then $L \not\sim_s L'$.

Proof. Let $R = \{(x, \bar{x}) : x \in S\} \cup \{(q, r)\}$. For every copied pair other than the source, direct moves are copied. At $(p, \bar{p})$, the original edge is matched by $\bar{p} \xrightarrow{a} r \xrightarrow{\tau} \bar{q}$, while $\bar{p} \xrightarrow{a} r$ is matched by $p \xrightarrow{a} q$ and ends in (q, r) . At (q, r) , moves from q are matched after $r \xrightarrow{\tau} \bar{q}$; that silent move is itself matched by zero silent moves at q . Hence R is a weak bisimulation, and repetition follows by transitivity. Under the additional conditions, a strong match for $\bar{p} \xrightarrow{a} r$ must relate r to an a -successor x of p , but $r \xrightarrow{\tau} \bar{q}$ would require the excluded direct τ -move from x . Thus strong bisimilarity fails. $\square$

Proposition 4 (Limit of label-blind structural comparison). *Let M be any comparator determined only by unlabeled structure and invariant under relabeling of transitions. Then M cannot characterize weak-trace equality, weak simulation or weak bisimilarity for labeled systems in general.*

Proof. Take two systems with the same two states and one directed edge. Label the edge a in L_a and $b \neq a$ in L_b . Their unlabeled structures are identical, so $M(L_a, L_b) = M(L_a, L_a)$. Yet their languages are $\{\epsilon, a\}$ and $\{\epsilon, b\}$, and neither observable move can be matched by the other system. Thus no decision based only on M can characterize the stated labeled relations. $\square$

Proposition 4 does not identify a defect in PN-GDDA. It states that a label-blind structural method answers a different question and cannot substitute for a labeled behavioral relation.

Example 1 (Equal traces, different branching). Let L_{late} have transitions

$$\ell_0 \xrightarrow{a} \ell_1, \quad \ell_1 \xrightarrow{b} \ell_b, \quad \ell_1 \xrightarrow{c} \ell_c, \quad (2.14)$$

and let L_{early} have

$$e_0 \xrightarrow{a} e_b, \quad e_0 \xrightarrow{a} e_c, \quad e_b \xrightarrow{b} e'_b, \quad e_c \xrightarrow{c} e'_c. \quad (2.15)$$

Both exact prefix-closed languages are $\{\epsilon, a, ab, ac\}$. Nevertheless, $L_{\text{late}} \not\sim_w L_{\text{early}}$. There are no silent moves, and matching either early a -successor with ℓ_1 leaves one of the two late actions unmatched. Directionally, $L_{\text{early}} \preceq_w L_{\text{late}}$: the late-choice state can match the selected b or c continuation of either early branch. The reverse $L_{\text{late}} \preceq_w L_{\text{early}}$ fails because no single early a -successor offers both continuations. Thus exact trace equality does not determine branching-time equivalence or simulation orientation.

2.7. Computational implementation

Algorithm 1 summarizes the executable path. Reachability and the formal relations are exact whenever the finite state space is completed below the declared state and token caps. The distance d_k , PN-GDDA, LTS-GDA, and the structural profile are diagnostics rather than formal relations.

Algorithm 1 (Executable comparison pipeline).

- (i) Enumerate the reachable LTS from the initial marking or logical state by a worklist; stop with an explicit error if a cap is exceeded.
- (ii) Cache $\text{Cl}_\tau(s)$ and construct $W_a(s)$ by collecting a -successors from the closure and the closures of their targets.
- (iii) Initialize $R = S_1 \times S_2$ and delete a pair immediately when it fails the selected transfer operator; repeat complete passes until none is deleted.
- (iv) Evaluate strong and weak bisimilarity and both simulation directions, apply Eq. (2.8), and attach trace and structural diagnostics.

Table 1. Controlled benchmark; the expected class is not used by the classifier.

Case	Change	Expected	Observed
Identity	exact copy	strong equivalence	strong equivalence
Silent refinement	insert internal steps	weak equivalence	weak equivalence
Label mismatch	replace an observable	not comparable	not comparable
Order swap	preserve counts, change order	not comparable	not comparable
Added branch	add an outcome	reference in candidate	reference in candidate
Early/late choice	move decision point	candidate in reference	candidate in reference

3. Computational implementation and formal/software verification

3.1. Construction-level software benchmark and comparator study

Six deterministic pairs were specified before any biological application. They provide behavioral-class ground truth by construction and test whether the software recovers intended formal distinctions; they do not provide biological ground truth. A 12-step labeled workflow supplies identity, controlled silent subdivision, label mismatch, label-order swap, and an added observable branch. The sixth pair is $a.(b+c)$ versus $a.b + a.c$ from Example 1. Table 1 gives construction-level and computed classes.

On this construction-level decision task, weak bisimulation recovered all six binary weak-equivalence decisions. Truncated trace equality recovered 5/6 but accepted the trace-equivalent branching pair; LTS-GDA recovered 4/6, the structural profile 3/6, and direct PN-GDDA 2/6. PN-GDDA assigned 1.0 to identity, silent refinement, label mismatch, and order swap, while added-branch and branching pairs scored 0.9676 and 0.9657. No threshold over the six PN-GDDA scores exceeded 4/6 because two negative controls tied the positive controls at 1.0. Figure 2 reports all predeclared decisions.

The structural profile averages state-count, edge-count, label-multiset, and out-degree-multiset similarities. The trace diagnostic declares a depth-8 match when $d_8 = 0$. PN-GDDA was implemented from its published definition using 151 directed bipartite graphlets and the 592 published/Holmes orbit slots.⁵ For orbit j ,

$$N_G^j(k) = \frac{D_G^j(k)/k}{\sum_{r>0} D_G^j(r)/r}, \quad A^j(G, H) = 1 - \frac{\left[\sum_k (N_G^j(k) - N_H^j(k))^2 \right]^{1/2}}{\sqrt{2}}, \quad (3.1)$$

and PN-GDDA is the mean A^j . Synthetic LTSs use an exact state-machine Petri net encoding; curated pairs use their native nets. The 0.9 threshold is a fixed diagnostic decision, not an equivalence relation. LTS-GDA instead combines four reachable-graph orbits with labeled edge, two-step, divergence, and convergence motifs.

The catalog audit recovered 151 graphlets and 592 slots from hash-verified

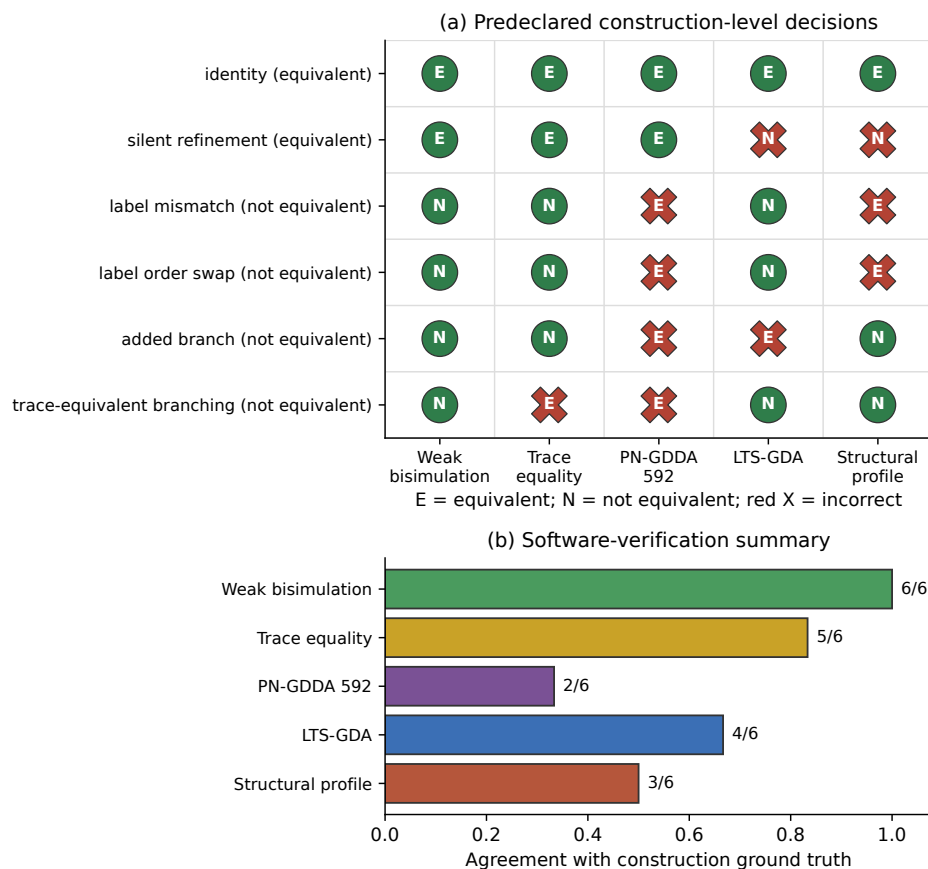
Controlled benchmark: software verification only

Fig. 2. Controlled software/formal benchmark. (a) Decisions for six cases whose labels were fixed by construction; a red cross marks disagreement with that label. (b) Agreement count on this limited task. Trace equality loses branching; PN-GDDA is label-blind; LTS-GDA retains local labels but does not abstract silent refinement or recursively match successors. The panel verifies implementation behavior and does not measure biological validity or generic method superiority.

Holmes 1.1.1/2.0.1.2 JARs. A direction- and type-preserving automorphism partition has 576 distinct orbits; using it is reported only as sensitivity. A Java probe matched every Holmes 1.1.1 topology/root assignment. On a four-node test net and its one-arc deletion, Python and Holmes scores were 0.9872027465870733 and 0.987202746587073, agreeing to 12 decimal places. Holmes is an independent software reference for the structural score.²¹

3.2. Independent formal oracles

Each analyzed LTS was exported in Aldebaran AUT format and evaluated with `ltscompare` from hash-pinned mCRL2 202607.0.²² The tool independently checked strong bisimilarity, weak bisimilarity, and exact weak-trace equivalence with `tau` hidden. Its used interface does not expose the weak simulation preorder, so no one-way simulation is attributed to mCRL2.

A separately coded attacker–defender game grows the least set of positions from which an unmatched move can be forced. It agreed with the production simulation algorithm on all 67,600 ordered pairs among all 260 one- and two-state LTSs over $\{a, \tau\}$. This is exhaustive within that finite universe, not a proof for arbitrary systems. Across the six synthetic pairs, six public-model controls, and nine HPN comparisons below, Python and mCRL2 agreed on all 42 reported strong/weak decisions. Theorem 1, finite exhaustive testing, and external-tool comparison are distinct evidence layers.

3.3. Observed computational scaling and reproducibility

Linear workflows of 8, 16, 32, 64, 96, and 128 transitions were compared with an identity and a silently refined copy. Median runtime over three repetitions rose from about 0.4 ms for nine states to about 0.6 s for the largest pairs. The 128-step identity relation had 16,641 candidate pairs; refinement increased it to 18,705 pairs over 129 and 145 states. Every class remained as constructed (the complete plot and table are archived in the repository). Timings used Python 3.11.8 on an Apple M3 Pro with 18 GiB memory and are machine-dependent.

The Python standard library implements reachability, formal relations, parsers, and controlled models. NumPy, pandas, Matplotlib, NetworkX, pydot, and Graphviz support tabulation and figures. Fixed seeds, source hashes, machine-readable outputs, unit tests, an executed notebook, environment files, and continuous integration are public. Runtime CSV values are excluded from byte-level regression because they depend on hardware; all scientific outcomes and other result tables are deterministic.

4. Biological network applications

4.1. Biological question and observational-interface principles

The central question is whether the animal/human and *Arabidopsis thaliana* DNA-damage models encode a common executable control organization, and at which observational resolution that correspondence ceases to hold. Independent biology supports comparing broad functions but also requires preserving mechanistic distinctions. ATM- and ATR-associated sensing, checkpoint control, repair, and recovery occur in both kingdoms; mammalian p53/apoptosis and the plant-specific SOG1, WEE1, SMR, and damage-associated endoreduplication programs are not interchangeable mechanisms.^{6–11, 23, 24}

Table 2. Biological rationale for the GIM observational interfaces. “A” denotes coarse function; “B/C” denotes the pathway-resolved interfaces.

Process	Animal/human encoding	<i>Arabidopsis thaliana</i> encoding	Interface treatment and independent support
Damage sensing	ATM-dominant DSB and ATR-dominant replication-stress channels	ATM/ATR channels upstream of SOG1	A merges detection; B/C distinguish the initiating channels. ^{7, 8, 11}
Checkpoint	CHK1/2-p53-p21 control	SOG1-regulated WEE1 and SMR control	A exposes function; B/C expose ATM/ATR but keep downstream relays internal. ^{6, 9, 10, 23}
Repair	DSB and excision-repair families followed by recovery	Broad DSB and excision-repair families followed by recovery	A merges repair; B/C separate repair families. ^{6, 8, 24}
Unresolved damage	Apoptotic exit from the proliferative pool	SMR-associated arrest followed by a combined differentiation/endoreduplication outcome	A/B collapse terminal loss; C distinguishes encoded mechanisms. ^{6, 10, 11, 25}

We therefore evaluated three biologically motivated interfaces on exactly the same two Petri-net structures and initial markings. Interface A observes five coarse functions: damage detection, checkpoint activation, repair, restoration, and exit from the proliferative cycle. Interface B additionally distinguishes double-strand-break and replication-stress detection, ATM- and ATR-associated signaling, and broad repair families, while retaining a common terminal `cell_cycle_exit` readout. Interface C preserves those distinctions and separately exposes mammalian `apoptosis`, plant `smr_induction`, and the plant net’s combined `differentiation_or_endoreduplication` transition. The last label is a property of this compact encoding: the model does not infer differentiation and endoreduplication as separate branches. Table 2 records why each process is observed or hidden. The executable definitions and the literature keys were fixed in source data and are exported as a machine-readable audit table.

4.2. *GIM: encoded DNA-damage control across observational resolutions*

The animal and plant nets reach 13 and 14 states, respectively. Because only transition labels change across interfaces, their native PN-GDDA structural similarity remains 0.9975846 (0.9975175 with the independently derived 576-orbit partition). This high value says that the selected Petri-net wiring is similar; it says nothing about labels, initial states, or reachable branching.

At Interface A the two LTSs are weakly, but not strongly, bisimilar: every displayed detection-checkpoint-repair path and each restoration or cycle-exit branch

Table 3. Executed GIM interface-sensitivity analysis. $A \preceq P$ means that the animal-model behavior is weakly simulated by the plant model; $P \preceq A$ is the reverse direction. Strong bisimulation was false for all three interfaces.

ID	Observational resolution	$A \preceq P$	$P \preceq A$	Formal class	d_6	PN-GDDA
A	Coarse functional/process	Y	Y	weak bisimulation	0.0000	0.9976
B	Damage/signaling/repair pathways; terminal fate collapsed	Y	Y	weak bisimulation	0.0000	0.9976
C	Mechanism-resolved terminal fate	N	N	not comparable	0.3529	0.9976

can be matched after internal relays are hidden. The result persists at Interface B, so separating damage, ATM/ATR, and repair channels does not by itself break the relation in these encodings. At Interface C, however, weak bisimilarity and both simulation directions fail. The truncated-trace distance changes from $d_6 = 0$ at A and B to $d_6 = 0.3529$ at C, while LTS-GDA falls from 0.9508 to 0.8921 (Table 3). The structural score is unchanged because no place, transition, or arc was altered.

The comparison thus contributes information unavailable from structure alone: it locates the observational boundary of the encoded correspondence. The A/B result is consistent with shared high-level damage-response functions, but does not identify p53 with SOG1 or plant terminal responses with apoptosis. The C result reflects the known distinction between these mechanisms and shows that the apparent correspondence depends on collapsing them into a common functional outcome. A limited model-derived hypothesis follows: assays or model readouts that aggregate terminal loss of proliferative capacity should be more compatible with the A/B correspondence, whereas readouts distinguishing apoptosis, SMR induction, and differentiation/endoreduplication should break it. This is a testable consequence of the two encodings, not an experimentally established cross-organism claim. Full Petri-net diagrams are retained in Appendix Appendix A.

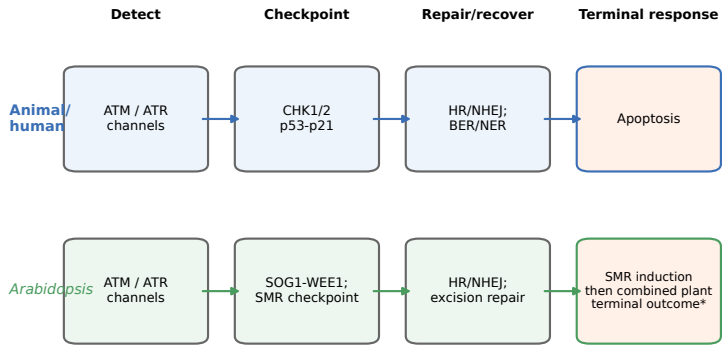
4.3. Cross-format ingestion and execution controls

Two externally authored GINsim models test the complete import-to-comparison path: the 13-component mammalian cell cycle in SBML-qual and the four-component multilevel p53-Mdm2 model in GINML.^{13–16} Files, URLs, and SHA-256 values are archived. Unitary-asynchronous updates move one enabled component by one level and label its direction. Each base model was compared with an exact copy, five silent subdivisions, and a reachable-label perturbation. All six controls recovered their predeclared strong/weak/exact-trace outcomes in Python and mCRL2 (Table 4).

Synchronous execution reached 13 of 826 asynchronous cell-cycle states (Jaccard 0.0157) and eight of 17 p53-Mdm2 states (Jaccard 0.471), confirming sensitivity to update semantics. These are cross-format execution controls, not biological validation.

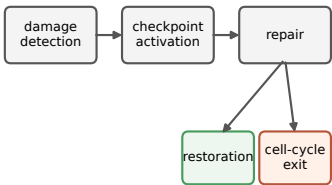
GIM: encoded functional correspondence depends on observational resolution

(a) Encoded DNA-damage control: shared functions, distinct implementations



*The encoded plant net combines differentiation/endoreduplication as one terminal outcome; the two are not separately inferred.

(b) Coarse interface: matched observable control flow



Organism-specific relays and terminal implementations are τ -internal.

Both encoded LTSs can match every displayed branch weakly.

(c) Interface refinement: executed result

ID	PN-GDDA	Formal class	d_6
A	0.9976	weak bisimulation	0.000
B	0.9976	weak bisimulation	0.000
C	0.9976	not comparable	0.353

A Shared functions; mechanisms hidden.

B Damage/ATM/ATR/repair exposed; terminal fate collapsed.

C Apoptosis, SMR, and the plant terminal outcome distinguished.

Fig. 3. Central GIM case study. (a) The encoded models share broad damage-response functions but use distinct regulatory and terminal mechanisms. (b) At the coarse interface, both reachable LTSs match the displayed branching after organism-specific relays are hidden. (c) Executed sensitivity analysis: Interfaces A and B are weakly bisimilar, whereas Interface C is non-comparable when terminal mechanisms are separate observations. PN-GDDA remains 0.9976 because the Petri-net structures do not change. These are model- and interface-conditional results, not organism-level equivalence.

Table 4. Public GINsim models and control outcomes. S/W/T denote strong, weak, and exact weak-trace equivalence.

Model	Format	States	Edges	Copy	Silent	Perturbed
Mammalian cell cycle	SBML-q	826	3,413	Y/Y/Y	N/Y/Y	N/N/N
p53-Mdm2 damage	GINML	17	26	Y/Y/Y	N/Y/Y	N/N/N

Table 5. HPN-DREAM families and structure-only representatives.

Cell line	Family	Medoid row ^a	Clauses	Mean Jaccard
BT20	72	24	29	0.815
BT549	191	137	25	0.854
MCF7	21	9	23	0.851

Note: ^aZero-based row; selected from model structure only.

4.4. *Exploratory HPN-DREAM/CASPOTS held-out analysis*

The public HPN-DREAM phosphoproteomic series and CASPOTS Boolean families contain 72 BT20, 191 BT549, and 21 MCF7 networks.^{17,18} One representative per cell line was selected before test responses were read: the model with maximum mean within-family Jaccard similarity between active clauses, with exact lexicographic tie-breaking. This structure-only rule selected zero-based rows 24, 137, and 9 (Table 5); an anti-leakage test forbids access to test files during selection.

The interface is the intersection of seven measured PI3K/MAPK/mTOR readouts. The three common test perturbations are mTOR inhibition, IGF1 plus mTOR inhibition, and 4EBP1 plus mTOR inhibition. Measured time-zero values initialize readouts; stimuli and inhibitors are clamped; other relevant nodes start at zero. Primary asynchronous LTSs label interface updates and hide all others. The nine cross-cell pair-condition comparisons were repeated synchronously; this stress test is not a most-permissive analysis. CASPOTS itself evaluates asynchronous trajectories.^{18,26}

Experimental distance is RMSE over common post-zero times and readouts. Spearman associations use an exact one-sided permutation of the three cell-pair labels within each condition ($3!^3 = 216$). CASPOTS/Clingo compatibility scores consume optimization to completion, use deterministic secondary tie-breaks, and were reproduced twice. Native specificity uses the exact sign permutation of three native-minus-mean-cross differences.

Selected LTSs had 2–44 states. None of the nine cross-cell pairs was weakly bisimilar: six had one-way simulation and three were not comparable. Direction was retained explicitly, with three left-model-in-right-model and three right-model-in-left-model relations. These categories do not form a justified one-dimensional order, and the held-out RMSE is symmetric between each cell pair. We therefore

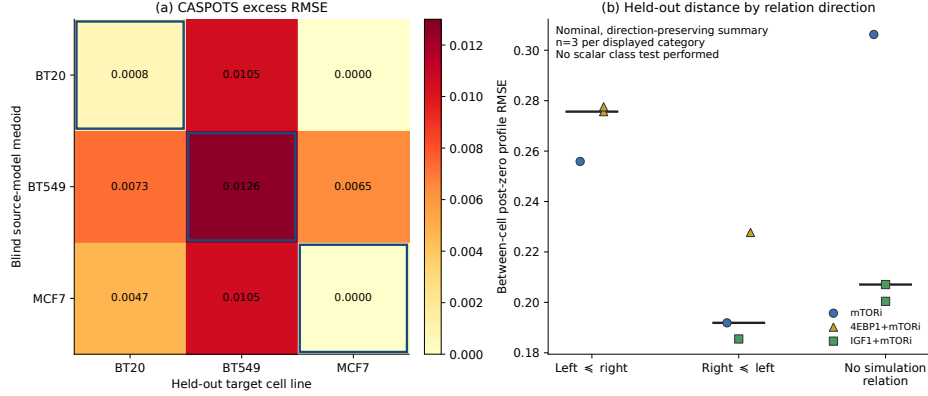


Fig. 4. Exploratory held-out HPN-DREAM analysis. (a) CASPOTS excess RMSE by source medoid and target; outlined cells are native contexts. (b) Between-cell profile RMSE with the direction of weak simulation retained. Horizontal bars are medians; each category has $n = 3$, and no scalar formal-class test was performed. Neither panel establishes native cell-line specificity.

removed the former ordinal class coding and performed no scalar formal-class test. Descriptively, median RMSE was 0.2418 for the six one-way relations and 0.2071 for the three non-comparable pairs; after preserving direction, the corresponding medians were 0.2757, 0.1919, and 0.2071, each based on only three observations. These summaries are not inferential evidence.

The two predeclared continuous diagnostics were retained. Truncated-trace distance gave $\rho = -0.596$ (exact one-sided $p = 0.556$), and LTS-GDA distance gave $\rho = 0.669$ ($p = 0.097$) under 216 condition-stratified permutations. Neither association is conclusive. Synchronous updating preserved seven of nine formal classes.

Native medoid RMSE was 0.33024 (BT20), 0.31335 (BT549), and 0.27726 (MCF7), against discretization RMSE 0.32944, 0.30079, and 0.27726. Native excess-RMSE advantage over mean cross-cell medoids was 0.00212 and not established by the exact sign test ($p = 0.25$; Fig. 4). The result demonstrates an exploratory held-out compatibility check and separates formal similarity from empirical response similarity; it does not demonstrate cell-line specificity or predictive discrimination.

4.5. *Secondary curated comparisons*

Five compact plant–animal pairs encode DNA repair/genome instability (GIM), energy metabolism (DCE), cell-cycle control (SPS), cell death/autophagy (RCD), and immune control (AID). Comparative studies and cancer-hallmark literature guided component selection; dedicated reviews informed plant DNA-damage, cell-cycle, and autophagy mechanisms.^{8,27–29} Transition-level provenance is machine-checked, which establishes where encoded assumptions came from and whether documentation is complete. It does not establish that the nets reproduce independent experiments. These manually curated models contain no kinetics, concentrations, tissue

Table 6. Supporting curated model-pair outputs. GIM is reported at Interface A; all relations and d_6 values are interface-conditional, whereas PN-GDDA is structural.

Module	Process	Formal class	d_6	PN-592	Orbit-576
GIM	DNA repair	weak bisimulation	0.000	0.9976	0.9975
DCE	energy metabolism	weak bisimulation	0.000	0.9969	0.9968
SPS	cell-cycle control	weak bisimulation	0.000	1.0000	1.0000
RCD	death/autophagy	plant in animal	0.143	1.0000	1.0000
AID	immune control	non-comparable	0.429	0.9907	0.9904

context, or independent experimental calibration.

The ten nets produced 5–14 reachable states. The GIM row in Table 6 reports Interface A; GIM, DCE, and SPS were weakly but not strongly bisimilar; RCD gave plant-in-animal one-way simulation; AID was non-comparable. Table 6 juxtaposes these classes with native PN-GDDA. The 592- and 576-orbit variants changed no score by 0.001 and no threshold decision: all five pairs were structurally similar at 0.9 despite three different behavioral classes.

RCD supplies a secondary directional result: both encoded nets contain stress, mTOR inhibition, autophagy, and survival, but only the animal encoding exposes a canonical caspase-apoptosis branch; the plant branch uses an internal metacaspase event. The plant LTS is weakly simulated by the animal LTS, whereas the reverse direction fails. This means containment only for the declared labels of these two models; it does not equate plant programmed cell death with mammalian apoptosis. Silent subdivision preserved every class over 300 refinements per module, while targeted relabeling changed the expected relations. Distances against 2,000 label permutations assess internal interface sensitivity, not external biological significance. DCE, SPS, and AID are retained as supporting model-ingestion and relation-diversity examples rather than independent biological validation.

5. Discussion

5.1. Structural and behavioral correspondence

The classifier turns an ordered pair of executable models into one statement at a declared observational interface. Proposition 2 supports the reporting priority: strong bisimilarity entails weak bisimilarity, and weak bisimilarity entails both simulation directions. One-way classes retain their orientation and therefore express behavioral containment, not equivalence. The fixed-point theorem concerns the algorithm actually executed, including its in-place deletion order and treatment of τ ; software tests complement that proof but cannot replace it.

The controlled cases show why structural, linear-time, and branching comparisons answer different questions. Identical unlabeled structure cannot determine labeled behavior (Proposition 4), while Example 1 has identical exact finite traces but different branching. PN-GDDA agreed with 2/6 construction-level behavioral labels even though the independent Holmes implementation reproduced every structural

score to 12 decimal places. This is not a defect in PN-GDDA: labels, initial states, and reachable branching are outside that statistic by design. Similarly, LTS-GDA retains labeled local motifs but does not perform silent abstraction or recursive successor matching.

GIM makes that complementarity biologically interpretable. Its structural score remains 0.9976 under all three interfaces, while the behavioral class changes from weak bisimulation to non-comparability only when terminal mechanisms become observable. The structural result describes similar selected wiring; the behavioral result describes what the reachable models can match under a biological readout. Neither statement subsumes the other.

5.2. *Interface and update semantics*

Every relation is indexed by Σ . Declaring an event internal changes the paths hidden by abstraction, while merging outcomes under one label coarsens the question. In GIM, A and B collapse apoptosis and the encoded plant terminal sequence into `cell_cycle_exit`; their weak bisimilarity consequently means agreement only at that functional resolution. Interface C separates those events and is non-comparable in both simulation directions. The interface is therefore a falsifiable modeling assumption, not a post-processing convenience. Proposition 3 covers the controlled transition subdivisions used here, but not refinements that introduce conflicts, interleavings, or observable choices.

Update semantics is likewise part of the comparison contract. Switching between asynchronous and synchronous execution changed public GINsim behavior and two HPN classes. General-asynchronous, most-permissive, and stochastic conventions were not evaluated.²⁶ A reported relation should therefore always name both its observable interface and execution semantics.

5.3. *Evidence and biological limits*

Table 7 separates four claims that are otherwise easy to conflate. Layer I is mathematical. Layer II asks whether the implementation computes the declared objects: Python and mCRL2 agreed on all 42 strong/weak decisions, and the independent game agreed on all 67,600 ordered pairs in its complete two-state universe. Layer III concerns the executable biological models. Hashes, parser controls, and transition provenance make ingestion and assumptions reproducible, but do not show that a model reproduces biology. Layer IV concerns independent biological support. The GIM interpretation is consistent with literature on shared functions and distinct mechanisms, but its interface-dependent relation remains derived from curated models; its assay consequence is a hypothesis. HPN supplies held-out measurements, but only nine pair-condition observations and no conclusive inferential result.

The HPN limitations are substantive rather than presentational. Directional simulation relations do not define an ordinal scale, so assigning them scalar ranks would discard their meaning; the revised analysis reports the directions nominally

Table 7. Evidence taxonomy used for claims in this study.

Layer	Evidence used here	What it supports	What it does not establish
I Formal	Definitions, propositions, fixed-point proof, finite termination	Correctness of the stated mathematics and executed relation operator	Correct software or biological relevance by itself
II Software	Six constructed cases, mCRL2, independent game, regression tests	Recovery of declared formal distinctions in tested domains	Biological usefulness or empirical validity
III Model	Hash-pinned GINsim files, parser controls, transition provenance, explicit semantics	Reproducible origin, ingestion, and assumptions of the encoded models	Experimental calibration or faithful organism-level behavior
IV Biological	Independent DDR literature; exploratory HPN-DREAM/CASPOTS responses	Biological plausibility and a route for confronting formal outputs with data	Confirmed cross-organism equivalence, prognosis, or cell-line specificity

and makes no class-level inferential test. The retained trace and LTS-GDA associations are inconclusive, and the CASPOTS native-versus-cross advantage is not established ($p = 0.25$). CASPOTS RMSE is minimum-distance compatibility over admissible trajectories, not a unique time-series prediction.¹⁸ These observations do not support prognostic or cell-line-specific claims.

The curated Petri nets additionally omit kinetics, concentrations, stochastic effects, tissue context, and independent experimental calibration. Provenance establishes why a transition was included and whether its documentation is complete; it is not model validation or validation of a biological conclusion. Label-permutation results measure sensitivity within those constructions, and cross-organism weak relations never imply organism-level functional identity.

5.4. Computational scope and use

The implementation explicitly enumerates states and may store $|S_1||S_2|$ candidate pairs. The scaling experiment reached 145 states and 18,705 pairs, so it does not establish genome-scale performance. Symbolic state representations, partition refinement, and compositional checking are natural extensions. PN-GDDA was applied to native curated nets and canonical state-machine encodings; using it on SBML-qual or Boolean rules would require a non-unique reverse translation. Larger repository models, prospectively expert-labeled relations, biological replicates, blinded interface selection, and additional update semantics are required before broader predictive claims. At present, the framework is best used to audit two explicit qualitative models at a predeclared observational resolution.

6. Conclusion

Comparing executable qualitative biological models requires an explicit account of observables, silent events, initial states, branching, and update semantics. The classifier reports the strongest supported relation from strong bisimilarity through directional weak simulation, with fixed-point correctness and termination established for finite LTSs. Controlled cases, mCRL2, and the independent game verify distinct parts of the implementation; they are not biological validation.

In the unchanged GIM Petri-net structures, PN-GDDA remained 0.9976 while the behavioral relation changed with observational resolution. Coarse function and pathway-resolved interfaces were weakly bisimilar, but exposing mammalian apoptosis and the encoded plant SMR-differentiation/endoreduplication sequence made the models non-comparable in both directions. The useful conclusion is not that behavioral comparison is generically superior to structural comparison, but that the two expose complementary properties and that an explicit interface locates where apparent model correspondence breaks. Public-model controls and the exploratory HPN analysis show how the audit crosses formats and can be confronted with data, while the inconclusive HPN statistics delimit current evidence. The framework supports reproducible, question-specific model auditing; it does not establish prognosis, cell-line specificity, mechanistic identity, or organism-level equivalence.

Appendix A. Detailed GIM Petri-net encodings

Figures 5 and 6 retain the full executable diagrams underlying the biological schematic and all three interface analyses. The place/transition structures and initial markings are fixed; interface-specific labels are assigned reproducibly by `gim_models_for_interface`.

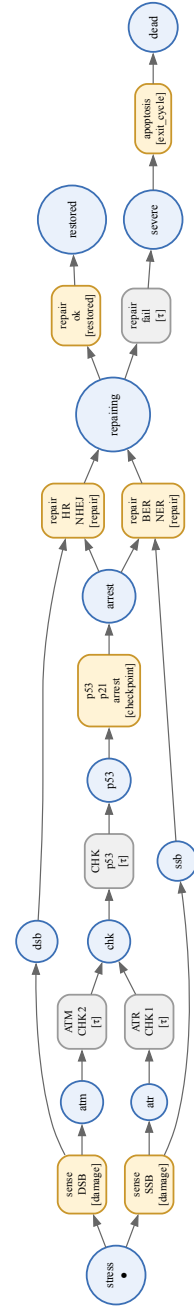


Fig. 5. Detailed animal/human GIM Petri net. Circles are places, rounded boxes are transitions, brackets show the base/coarse observable labels, and gray τ transitions are internal.

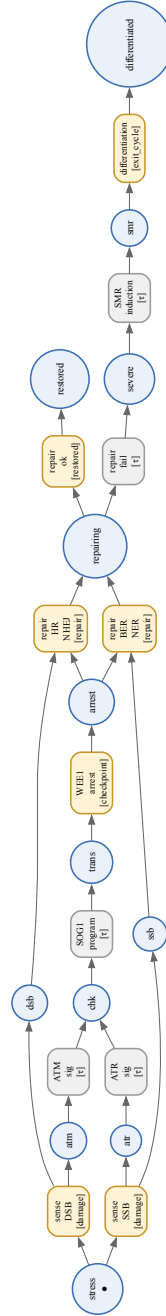


Fig. 6. Detailed *Arabidopsis thaliana* GIM Petri net, using the same graphical conventions as Fig. 5. The terminal transition is a combined outcome in this encoding and does not distinguish differentiation from endoreduplication.

Data and code availability

Code, curated definitions, hash-verified public models, normalized response files, metadata, and generated results are available under the MIT License at <https://github.com/cero1979/BisimulationMol>. The repository documents formal/software verification, interface definitions, figure regeneration, tests, and notebook execution. Machine-readable GIM and HPN results are provided as `results/gim_interface_sensitivity.csv` and `results/hpn_dream_formal_data_validation.csv`. No confidential or participant-level data were used.

Declarations

Funding. No specific funding was received. **Competing interests.** The author declares no competing interests. **Ethics.** Only computational models and published aggregate data were used; no human participants, samples, or live animals were involved. **Author contributions.** Carlos Ramirez Ovalle performed the conceptualization, methodology, software, formal analysis, validation, visualization, drafting, and revision.

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